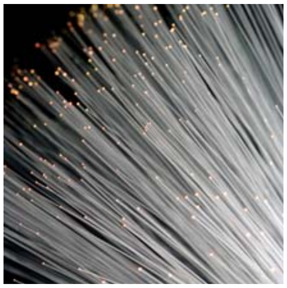
A 2.3.2 Transmission Media



A2.3.2 Compare types of media for data transmission.

Data travels in diverse ways, from fibre optics to wireless signals. Let's compare wired (fiber, twisted pair) and wireless transmission, weighing factors like bandwidth, cost, and security to understand their unique strengths and weaknesses.

Introduction

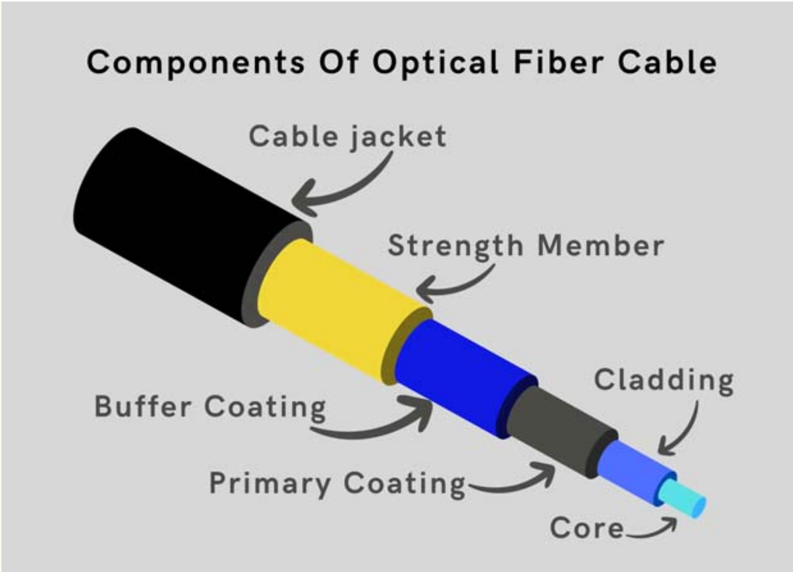
The Internet and modern communication systems transmit data using various media. Understanding the characteristics of these media is crucial for designing and optimising networks. This section explores the key differences between wired (fibre optic and twisted pair) and wireless transmission, examining their advantages and disadvantages.

Wired Transmission: Fibre Optic and Twisted Pair Cables

Fibre Optic Cables

Fibre optic cables transmit data using light pulses through thin strands of glass or plastic. These cables offer extremely high bandwidth and are ideal for long-distance communication and high-speed internet connections.

How it works: Light signals are generated by lasers or LEDs and travel through the fibre's core, reflecting off the cladding (outer layer) to minimise signal loss.



A fibre optic cable cross-section diagram showing the core, cladding, and protective jacket.

Twisted Pair Cables:

Twisted pair cables consist of copper wires twisted together to reduce electromagnetic interference. These cables are commonly used in Ethernet networks and telephone lines.

How it works: Electrical signals are transmitted through the copper wires. The twisting of the wires helps cancel out interference from external sources and other pairs within the cable.

A close-up image of a twisted pair cable showing the twisted wires and insulation. **Source: Wikimedia**

Wireless Transmission

Wireless transmission uses electromagnetic waves to transmit data through the air. This includes technologies like Wi-Fi, cellular networks, and satellite communication.

How it works: Data is encoded into electromagnetic waves transmitted through the air. Antennas receive these waves and decode them back into data.

Advantages and Disadvantages of Each Transmission Type

Feature	Fibre Optic	Twisted Pair	Wireless
Bandwidth	Extremely high	Moderate	Variable
Range	Long	Short to medium	Variable
Susceptibility to Interference	Low	Moderate	High
Attenuation	Low	Moderate to high	High
Reliability	High	Moderate	Variable
Security	High	Moderate	Variable (can be low)
Cost	High (installation)	Low	Variable

Feature	Fibre Optic	Twisted Pair	Wireless
Installation Complexity	High	Moderate	Low to moderate

Fibre Optic:

Advantages: High bandwidth, low attenuation, high security, immune to electromagnetic interference.
Disadvantages: High installation cost, fragile cables, complex installation.

Twisted Pair:

Advantages: Low cost, easy installation, widely available.
Disadvantages: Limited bandwidth, susceptible to interference, high attenuation over long distances.

Wireless:

Advantages: Mobility, ease of installation, flexibility.
Disadvantages: Limited bandwidth, susceptible to interference, security risks, variable reliability.

Factors to Consider: Bandwidth, Complexity, Cost, Range, Interference, Attenuation, Reliability, Security

Bandwidth: Fibre optic offers the highest bandwidth, followed by twisted pair, and then wireless, which varies greatly.
Complexity of Installation: Wireless is generally the easiest type of installation, followed by twisted pair and fibre optic, which require specialised equipment and expertise.
Cost: Twisted pair is the cheapest, followed by wireless (depending on the technology), and then fibre optic, which has high installation costs.
Range: Fibre optics can transmit data over the longest distances, followed by wireless (depending on the technology), and then twisted pair, which is limited to shorter distances.
Susceptibility to Interference: Fibre optic is immune to electromagnetic interference, twisted pair is moderately susceptible, and wireless is highly vulnerable.
Attenuation: Fibre optic has the lowest attenuation, twisted pair has moderate to high attenuation, and wireless has high attenuation.
Reliability: Fibre optic is the most reliable, followed by twisted pair, and then wireless, which can be affected by interference and signal strength.

Security: Fibre optic is the most secure, twisted pair is moderately secure, and wireless requires strong encryption to be secure.

Key questions

Q: Why is fibre optic cable used for long-distance communication?

A: Fibre optic cables have low attenuation, allowing signals to travel long distances with minimal loss. They also offer high bandwidth, which is necessary for transmitting large amounts of data.

Q: What is the primary advantage of using twisted pair cables in Ethernet networks?

A: Twisted pair cables are inexpensive and easy to install, making them suitable for local area networks (LANs).

Q: Why is wireless transmission susceptible to interference?

A: Wireless transmission uses electromagnetic waves, which can be affected by other electromagnetic signals in the environment, such as those from other devices or atmospheric conditions.

Q: How does attenuation affect data transmission?

A: Attenuation is the loss of signal strength over distance. High attenuation can lead to data loss and reduced transmission speed.

Q: Why is security a concern with wireless networks?

A: Wireless signals can be intercepted by unauthorised users, making it easier to eavesdrop on data transmissions. Encryption is essential to protect wireless networks.

Q: When would twisted pair be preferred over fibre optic?

A: For short distances and when cost is a major factor. For example, in a small office network.

Q: What are some factors that influence the range of a wifi signal?

A: Obstacles like walls, other electronic devices, and the power of the wireless access point.

Q: How does bandwidth relate to data transmission speed?

A: Bandwidth is the capacity of a transmission medium to carry data. Higher bandwidth allows for faster data transmission speeds.

A 2.3.2 Quiz

Test your knowledge of media for data transmission

1

Wireless transmission | Reliability | Security | Interference | Twisted pair cables | Cost | Bandwidth | Attenuation | Fibre optic cables | Range

[**Fibre optic cables**] transmit data using light pulses through thin strands of glass or plastic.

[**Twisted pair cables**] consist of pairs of copper wires twisted together to reduce electromagnetic interference.

[**Wireless transmission**] uses electromagnetic waves to transmit data through the air.

[**Bandwidth**] is the capacity of a transmission medium to carry data.

[**Attenuation**] is the loss of signal strength over distance.

[**Interference**] is the disruption of a signal caused by external factors.

[**Security**] refers to the protection of data from unauthorized access.

[**Cost**] is a major factor to consider when choosing a transmission medium.

[**Range**] is the distance a signal can travel without significant loss of quality.

[**Reliability**] is the ability of a transmission medium to consistently deliver data without errors.

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